Fiscal Year:	FY 2004	**Task Last Updated:**	FY 02/03/2005
PI Name:	Pohorille, Andrew Ph.D.		
Project Title:	Computational Techniques for Reconstruction and Discovery of Metabolic, Signal Transduction and Evolutionary Pathways		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035	**Congressional District:**	14
Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	07/01/2001	**End Date:**	06/30/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Shrager, Jeff (Institute for Studying Learning and Expertise) Langley, Pat (Institute for Studying Learning and Expertise)		
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We developing computational tools, collectively called BioLingua, that aid biologists in forming and evaluating explanations of cellular processes in a semi-automated manner. These tools are particularly suitable for reconstruction and discovery of metabolic, signaling, and evolutionary pathways in biological systems. They are necessary if biologists are to take full advantage of the newly acquired ability to collect large amounts of data through genomic, proteomic and microarray studies. One specific area of application of BioLingua is help predicting behavior of organisms in response to space environement, such as microgravity and radiation. Task objectives include developing: 1 A formalism and programming language for expressing models of biological processes at varying levels of abstraction, capable of supporting both qualitative and quantitative descriptions. 2 Simulation software that operates on this formalism to make qualitative and quantitative predictions about the behavior of a biological system under specific environmental conditions. 3 Explanatory software that generates hypotheses about metabolic and gene regulation pathways and networks, evaluates them in the context of experimental data collected by the biologist and prior knowledge, and proposes revisions to the models to improve their agreement with both the data and prior knowledge. Progress in the areas of genomics and proteomics leads the way to revolutionary changes in biological sciences. The key issue that remains unresolved is how to connect the information emerging from these areas with the behavior of whole biological systems, such as cells, organism or ecosystems. Microarray technologies provide the means to measure levels of expression for tens of thousands of genes in a single experiment. Similarly, the constantly improving methods of proteomics allow for measuring levels of many proteins. In principle, these data contain a wealth of information about the whole organism. To take full advantage of this information, it has to be combined with the rich biological knowledge that has already been accumulated and stored in the existing databases and knowledge bases. In practice, however, this task is so difficult that biologists, lacking adequate tools, usually limit themselves to analyzing a small subset of genes or gene products, comprising for example a single pathway. Thus, much of the data remain not interpreted. This problem is particularly relevant in NASA applications. For example, genomic flight experiments are rare and expensive. Therefore, it is of utmost interest to extract maximum information from the results. In addition, the effects of gravity on organisms are expected to be complex and indirect. Most likely, they cannot be understood by considering a single pathway or a small section of the regulatory network. Instead, complex analyses of regulatory and metabolic networks, and thorough comparisons between different organisms and different experimental conditions are likely to be needed. The computational approaches and tools that are being developed in this task make these complex analyses feasible and, by doing so, provide considerable added value to space and ground based biological experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The computational environment developed in this task is being used by biological, biomedical and pharmaceutical communities to understand genetic basis of diseases and develop effective and safe cures.
Task Progress:	A BioLingua application for the cyanobacteria is in Beta use worldwide. The principal causal model inference tools, which form the heart of the system have been tested in this domain, and in domains of bone development. BioLingua has been also extensively tested to examine the robustness and efficiency of the algorithm. Recently, BioLingua has been deployed at the NASA Center of Bioinformatics, Stanford Genome Technology Center and several other academic institutions..
Bibliography Type:	Description: (Last Updated: 04/09/2008)
Articles in Peer-reviewed Journals	Bay SD, Shrager J, Pohorille A, Langley P. "Revising regulatory networks: from expression data to causal models." J Biomed Inform. 2002 Oct-Dec;35(5-6):289-97. PMID: 12968777 , Jan-2003
Articles in Peer-reviewed Journals	Massar JP, Travers M, Elhai J, Shrager J. "BioLingua: a programmable knowledge environment for biologists." Bioinformatics. 2005 Jan 15;21(2):199-207. Epub 2004 Aug 12. PMID: 15308539 , Jan-2005